

MethylDetector (Theoretical foundation)

COMPUTATIONAL TEAM - EPI METHYL ANALYTICS

Summary

MethylDetector is a Python-based tool for detecting differentially methylated positions (DMPs) in genomic data using an enhanced centroid-based approach. It compares methylation profiles from two groups (stored as “centroids” in HDF5 format) to identify positions where methylation distributions differ significantly.

The statistical framework includes:

- Likelihood Ratio Tests (LRT) for Beta-distributed methylation rates,
- Normal approximations for faster computation when assumptions are held,
- Parameter estimation via Maximum Likelihood Estimation (MLE) or Method of Moments (MoM),
- False discovery rate (FDR) correction using Storey’s method,
- Global significance assessment with Stouffer’s method, and
- Grouping significant positions into contiguous genomic regions.

This report details the mathematical formulations, assumptions, and implementation of these methods. It draws directly from the project’s core modules:

- `detector.py` — main analysis class,
- `centroid_detector.py` — centroid comparison orchestration,
- `methylation_fdr_analysis.py` — FDR and global significance analysis,
- `comparer.py` — statistical computations with GPU acceleration.

1. Introduction

In DNA methylation studies, the amount of methylation at a specific genomic position can vary between individuals. These methylation levels are measured as fractions between 0 and 1. Because of this, we need a statistical model that works well with proportions. The Beta distribution is ideal for this task because it can describe a wide range of shapes—whether methylation is mostly low, mostly high, or somewhere in between.

Methylation data in genomics often follows a Beta distribution, parameterized by shape parameters α and β . The observed methylation level for an individual sample, denoted x_i , is treated as a draw from this distribution: $x_i \sim \text{Beta}(\alpha, \beta)$. This reflects that methylation levels at a genomic position are continuous proportions bounded between 0 and 1, with shapes that can vary from uniform to highly skewed depending on the degree of methylation across samples. The Beta distribution is flexible enough to capture this variety, modeling both symmetric and asymmetric patterns, as well as intermediate, hypomethylated, and hypermethylated states (Johnson NL, 1995) (Sanchez, 2023). We therefore treat across-sample methylation fractions at each position as $\text{Beta}(\alpha, \beta)$ with parameters varying by position and verify adequacy via goodness-of-fit and overdispersion checks.

For each genomic position, a **centroid** is a summary of methylation level statistics across all samples in a group, storing values such as the mean methylation, variance, sample size N , and sufficient statistics (e.g., sums of log-transformed methylation levels). These summaries preserve all the information needed to fit a Beta distribution for that position while avoiding the storage and processing cost of raw per-sample data, enabling fast and direct group comparisons (Feng H, 2014).

Sufficient statistics are quantities that capture all the information needed to estimate the parameters of a probability distribution from the data, without requiring access to the raw observations. For the Beta distribution, the sufficient statistics for a set of methylation fractions $\{x_1, x_2, \dots, x_N\}$ are the sum of the logarithms $\sum \log x_i$, the sum of the logarithms of the complements $\sum \log(1 - x_i)$, and the sample size N . These summaries are mathematically equivalent to the full dataset in terms of parameter estimation, enabling efficient storage and computation while preserving the ability to infer the shape parameters α and β .

The enhanced centroid-based approach in **MethylDetector** compares two centroids—each summarizing group-level methylation statistics for the same genomic positions—through a structured sequence of steps:

- **Identifying common positions.** Ensures both groups are compared only at genomic positions measured in all samples, so that Beta distribution parameters are estimated from directly comparable methylation level data.
- **Filtering positions with sufficient sample sizes** ($N \geq \text{min_N}$, default 10) . Positions with very small N yield unstable Beta parameter estimates and can produce misleading p-values; enforcing a minimum sample size improves statistical reliability and reduces false positives (Feng H, 2014).
- **Performing statistical tests to compute p-values for differences in Beta parameters.** Compares the fitted Beta distributions for the two groups at each position, using likelihood-based or approximation methods, to detect significant differences in methylation level distributions (Feng H, 2014).

GPU acceleration (via `CuPy`) is used for batch processing when available, falling back to CPU (`NumPy/SciPy`) otherwise.

2. Centroid Data Structure and Preprocessing

To speed up computation and reduce memory usage, we avoid working with raw methylation values. Instead, we summarize the data into "centroids," which are compact statistical summaries of methylation levels for each genomic position. These summaries keep only the information we need—like means, variances, and log values—to fit models and compare groups efficiently.

- Centroids are stored in `HDF5` files, each containing datasets that summarize methylation-level information for multiple genomic positions:
- **pos:** Genomic coordinates of the positions being analyzed.
- **N:** Number of samples contributing to the summary at each position.
- **mC, uC:** Total methylated and unmethylated counts across samples (used for quality control and optional count-based analyses).
- **Sml, Sml2:** Sufficient statistics for moment-based estimation — **Sml** stores the sum of methylation levels, and **Sml2** stores the sum of squared methylation levels. Together, they allow computation of the sample mean and variance without retaining the full vector of individual methylation values.
- **log_x_sum, log_1_minus_x_sum:** Logarithmic sums of methylation levels and of one minus methylation levels, respectively. These are the *sufficient statistics* for maximum likelihood estimation (MLE) of Beta distribution parameters (α, β). Storing

them precomputed makes it possible to fit models and run likelihood-ratio tests directly from the centroid data, without re-reading raw samples.

Common genomic positions between groups are identified using set intersection on the `pos` dataset. Positions are then filtered to remove invalid or unreliable entries — for example, those containing `NaN` or infinite values, or where the sample size `N` is below the configured threshold (`min_N`). This ensures that downstream p-value computations are based on stable and comparable statistical estimates.

Sufficient statistics are essential because they **condense all the necessary information for statistical inference into a small set of aggregate values**, which drastically reduces memory requirements and speeds up computation, especially when processing millions of genomic positions (Johnson NL, 1995) (Feng H, 2014) (Korthauer K, 2019).

3. Statistical Tests for Differential Methylation

To find out if methylation patterns differ between two groups (e.g., healthy vs. sick), we test each genomic position to see if the distributions are significantly different. These tests help identify where meaningful biological changes might be happening.

For each position, the null hypothesis H_0 is that the methylation distributions are identical (same Beta parameters), vs. H_1 that they differ.

3.1 Likelihood Ratio Test (LRT)

The likelihood ratio test (LRT) checks whether two groups are best described by the same methylation pattern or by two separate ones. If using one model for both groups fits worse than using two, we conclude that the methylation levels differ at that position.

The LRT compares the likelihood of data under separate Beta models (H_1) for each group vs. a combined model (H_0) shared by both groups (Wilks, 1938), in order to test whether the methylation level distributions at a given genomic position differ significantly between groups. Throughout, per-position across-sample methylation is modeled as $Beta(\alpha, \beta)$; tests compare group-specific Beta fits to a pooled model.

Beta Distribution and Log-Likelihood

The Beta PDF is: (Johnson NL, 1995)

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1 - x)^{\beta-1}$$

For N i.i.d. observations, $x_i \sim \text{Beta}(\alpha, \beta)$, the log-likelihood is:

$$\ell(\alpha, \beta) = n(\log \Gamma(\alpha + \beta) - \log \Gamma(\alpha) - \log \Gamma(\beta)) + (\alpha - 1) \sum \log x_i + (\beta - 1) \sum \log(1 - x_i)$$

Purpose: Writing the likelihood in this form makes it possible to estimate the Beta parameters (α, β) for each group and compare models directly via their maximum log-likelihood values, which is the basis of the LRT.

Why precomputed sums are used: In centroids, quantities such as $\log_x_sum = \sum \log x_i$ and $\log_1_minus_x_sum = \sum \log(1 - x_i)$ are stored as sufficient statistics (Feng H, 2014). This means the log-likelihood can be evaluated without loading every individual methylation value, saving both time and memory — a critical advantage when processing millions of positions.

Two-Group Comparison:

- Separate models: $\ell_1(\alpha_1, \beta_1) + \ell_2(\alpha_2, \beta_2)$
- Combined model: $\ell_c(\alpha_c, \beta_c)$ where all samples are pooled.

The LRT statistic is:

$$\Lambda = -2(\ell_c - (\ell_1 + \ell_2)) = 2[(\ell_1 + \ell_2) - \ell_c]$$

Under H_0 , Λ follows a χ^2 distribution with $df = 2$ asymptotically ($\Lambda \sim \chi^2(2)$), since H_1 has two more parameters than H_0 . (Wilks, 1938) The p-value is computed as:

$$p = 1 - F_{\chi^2}(\Lambda; df = 2)$$

Implementation Notes

- Uses digamma (ψ) and trigamma (ψ_1) functions for derivatives in MLE (Johnson NL, 1995).
- The digamma function (ψ) is the derivative of $\ln \Gamma(x)$ and appears in the gradient of the Beta log-likelihood with respect to α and β .

$$\psi(x) = \frac{d}{dx} \ln \Gamma(x) = \frac{\Gamma'(x)}{\Gamma(x)}$$

- The trigamma function, the derivative of the digamma, appears in the Hessian (second derivatives) used in Newton–Raphson optimization.

$$\psi_1 = \frac{d}{dx} \psi(x)$$

These special functions enable accurate and efficient estimation of α and β , even for large datasets.

- Includes fallbacks to handle numerical instability (e.g., when means are very close to 0 or 1).
- GPU version uses CuPy's special functions (`gpu_digamma`, `gpu_trigamma`) for accelerated computation when available.

3.2 Normal Approximation

When we have enough samples and the data behaves nicely, we can speed things up by using a normal approximation. Instead of fitting complex Beta distributions, we approximate them with simpler normal ones and run standard Z-tests.

For large N ($\geq \text{normal_approx_threshold}$, default 30) and well-behaved within the interior of the parameter space ($\alpha, \beta \geq \text{min_beta_params}$, default 2.0), the Beta distribution can be well-approximated by a normal distribution with:

$$\mu = \frac{\alpha}{\alpha + \beta}, \quad \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

$$\text{Beta}(\alpha, \beta) \approx \text{Normal}(\mu, \sigma^2)$$

This approximation allows the difference in means between two groups (μ_1 vs. μ_2) to be tested using a standard two-sample Z-test:

$$Z = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_2^2}{n_2}}}$$

The p-value is computed as:

$$p = 1 - \phi(Z)$$

where ϕ is the standard normal cumulative distribution function (CDF).

This approach is faster than the Likelihood Ratio Test (LRT) and is applied when the approximation conditions are met; otherwise, the analysis defaults to the LRT for accuracy (Johnson NL, 1995).

4. Beta Parameter Estimation

To compare methylation patterns, we need to estimate the shape of the Beta distribution for each group. This section shows two ways to do it: a quick and simple technique (Method of Moments, MoM), and a more accurate but slower one (Maximum Likelihood Estimation, MLE). Parameters α and β are estimated using either the MoM (default) or MLE for greater precision in edge cases.

4.1 Method of Moments (MoM)

MoM is a fast way to estimate distribution parameters using averages and variances. It is less precise than other methods but works well in most cases and is used as a fallback when needed.

The sample mean is computed as:

$$\mu = \frac{Sml}{N}$$

The unbiased sample variance of methylation levels is:

$$s^2 = \frac{Sml^2 - \frac{Sml^2}{N}}{N - 1}$$

Using these, the Beta parameters are:

$$k = \frac{\mu(1 - \mu)}{s^2} - 1$$

$$\alpha = \mu k, \quad \beta = (1 - \mu)k$$

Values are clipped to `min_beta_params` to prevent numerical instability. MoM is also used as a fallback when MLE fails to converge.

4.2 Maximum Likelihood Estimation (MLE)

MLE is a more accurate method that tries to find the parameter values that make the observed data most likely. It is more computationally intensive, but it gives better results—especially when the methylation values are extreme.

The MLE approach maximizes the Beta log-likelihood via **Newton's method**: (Johnson NL, 1995)

$$(\alpha, \beta)^{(k+1)} = (\alpha, \beta)^{(k)} - H^{-1} \nabla \ell$$

where:

- $\nabla \ell$ is the gradient, computed using the digamma (ψ) function.
- H is the Hessian, computed using the trigamma (ψ_1) function.

Procedure:

- Initial guess: Parameters from MoM.
- Iteration limit: `max_newton_iters` (default 50).
- Convergence tolerance: `newton_tol` (default 1×10^{-6}).
- For likelihood-ratio tests, maximum likelihood estimation (MLE) is attempted by default, with fallback to method of moments (MoM) when MLE fails to converge.
- GPU acceleration: Vectorized Newton's method over batches for speed.

For combined model in LRT, statistics are pooled (e.g., $mC_c = mC_1 + mC_2$).

5. Multiple Testing Correction: FDR with Storey's Method

When we test thousands or millions of genomic positions, some will show significant differences just by chance. If we just look at raw p-values, we will get many false positives. To avoid too many false positives, we adjust the results using Storey's method, which controls the false discovery rate (FDR) (`apply_fdr_correction=True`, `default method="storey"`).

5.1 Storey's q-Values

Storey's method estimates how many of the tests are truly null (i.e., no real difference in methylation), and uses that estimate to adjust the p-values into more reliable q-values (FDR-adjusted p-values).

How does it work?

1. Start with all p-values (one per position).
2. Estimate π_0 , the proportion of tests that are truly null:
 - Storey observes that large p-values (e.g., 0.5 to 0.95) are likely from null hypotheses.
 - It uses a tuning parameter λ to count how many p-values fall above λ , and estimates $\hat{\pi}_0$ as: (Storey, 2002) (Storey JD, 2003)

$$\hat{\pi}_0(\lambda) = \frac{\#\{p_i > \lambda\}}{m(1 - \lambda)}$$

where:

- p_i = p-value for the i -th tested position,
- m = total number of positions tested,
- λ = tuning parameter in $[0.01, 0.95]$

The estimate $\pi_0 = \min_{\lambda} \hat{\pi}_0(\lambda)$ is taken as the minimum $\hat{\pi}_0(\lambda)$ over the chosen λ values, capped at 1.

Given sorted p-values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$, q-values are computed as:

$$q_{(i)} = \min_{j \geq i} \frac{\hat{\pi}_0 m p_{(j)}}{j}$$

A position is considered significant if $q_{(i)} \leq \alpha$ (default $\alpha = 0.05$).

If an output directory is specified, the implementation also generates an interactive Plotly plot of $\hat{\pi}_0$ versus λ . If the correction step fails, the analysis falls back to uncorrected p-values.

6. Global Significance Analysis: Stouffer's Method

Instead of looking at each position separately, Stouffer's method combines all test results into a single overall score. This tells us whether, on the whole, there is a statistically significant difference between the two groups.

This step assesses the overall difference between centroids by combining the adjusted significance values (q-values, or raw p-values if no correction is applied) into a global test statistic.

The **Stouffer's Z-score** is computed as: (Whitlock MC., 2005)

$$Z = \frac{\sum_{i=1}^m w_i \phi^{-1}(1 - p_i)}{\sqrt{\sum_{i=1}^m w_i^2}}$$

where:

- m = total number of positions included in the global analysis,
- p_i = p-value at position i ,
- w_i = optional weight for position i (default: all weights equal to 1),
- ϕ^{-1} = inverse of the standard normal cumulative distribution function (CDF).

The global p-value is then:

$$p_{global} = 1 - \phi(|Z|)$$

Where ϕ is the standard normal CDF.

To maintain numerical stability:

- p-values are clipped to $[10^{-10}, 1-10^{-10}]$ before transformation to Z-scores to avoid infinite values.
- Extremely large or small Z-scores ($|Z| > 10$) are assigned a global p-value of 0, $p_{global} = 0$.

A global difference is considered significant if:

$$p_{global} \leq \text{global_significant_threshold} \quad (\text{default} = 0.05)$$

7. Grouping Significant Positions

Sometimes, nearby significant positions on the genome appear close together due to local biological changes. Instead of analyzing each one in isolation, we merge nearby positions into contiguous regions. This helps us identify broader patterns and makes downstream interpretation easier.

Significant positions often appear close together on the genome. This section explains how we merge nearby significant positions into larger regions to highlight broader changes in methylation patterns.

Procedure:

1. Sort the significant positions by genomic coordinate.
2. Merge adjacent positions into the same region if:

$$pos_{k+1} - pos_k \leq \text{max_gap}$$

3. **Output** the resulting regions as (start, end) coordinate tuples in a CSV file for downstream analysis.
- Merging is performed on the sorted position list and uses inclusive boundaries so that positions exactly at the `max_gap` distance are included in the same region. If data span multiple chromosomes, positions should be grouped by chromosome before this step.

8. Implementation Details and Performance

MethylDetector is built for scale. This section describes how it uses GPUs when available, falls back to CPU methods when needed, and includes safeguards for edge cases to ensure robust results across a wide variety of datasets.

- **GPU/CPU Switching:** Data is processed in batches on the GPU when available (`batch_size ≥ 100,000`). If a GPU is not available or the batch size is smaller, computation falls back to CPU processing with multiprocessing (`n_jobs=auto`).
- **Edge Cases:** Handles scenarios with small sample sizes (N), extreme parameter values (e.g. $\alpha < 1$ or $\beta < 1$) or means near 0 or 1. Fallback strategies include using the Method of Moments (MoM) when MLE fails and parameter clipping to `min_beta_params` to avoid divergence.
- **Outputs:** Generates multiple output formats for flexibility:
 - **JSON** files containing summary statistics,
 - **CSV** files for significant positions and grouped regions,
 - **TEXT** summary report,
 - **HTML** interactive plots for visual inspection.
- **Version:** Current release: `v0.2.0` — implementing the centroid-based approach for efficient large-scale methylation analysis.

9. Assumptions and Limitations

Like all statistical models, ours relies on some assumptions—about data distribution, independence, and sample size. Understanding these helps interpret results correctly and recognize when they might be overly conservative.

- **Distributional Assumption:** Methylation rates at each genomic position are assumed to follow a Beta distribution and to be independent across positions. Because methylation fractions are proportions in $[0,1]$ and vary across individuals, we model the across-sample distribution at each position as $x_i \sim \text{Beta}(\alpha, \beta)$. The Beta family is flexible (unimodal/skew/U-shaped), conjugate to the binomial sampling model, and empirically fits position-wise distributions in many datasets. We validated adequacy via goodness-of-fit checks and overdispersion diagnostics, *parameters are allowed to vary by position*. This assumption aligns with thermodynamic models of DNA methylation, where

spontaneous methylation changes follow a generalized gamma distribution derived from maximum entropy principles. (Sanchez, 2023)

- **Asymptotic Approximations:** The χ^2 reference distribution for the LRT and the normal approximation for the Z-test rely on large-sample theory; their accuracy may degrade when sample size N is small.
- **Multiple Testing and Global Tests:** Storey's FDR procedure assumes that some hypotheses are truly null. Stouffer's method assumes independence among tests; when tests are positively correlated, it tends to be conservative, potentially reducing power.
- **Environment Restrictions:** The implementation runs without internet access or package installation and is limited to libraries already available in the execution environment.

Conclusion

In summary, MethylDetector brings together speed and accuracy to identify significant methylation changes. Its statistical pipeline balances rigor with practical usability, making it well-suited for large-scale genomic studies.

MethylDetector's statistical pipeline enables efficient and robust detection of differentially methylated positions (DMPs), incorporating both multiple testing correction and global inference to ensure reliable results. The centroid-based design and GPU/CPU hybrid processing allow large-scale methylation datasets to be analyzed quickly while maintaining statistical rigor.

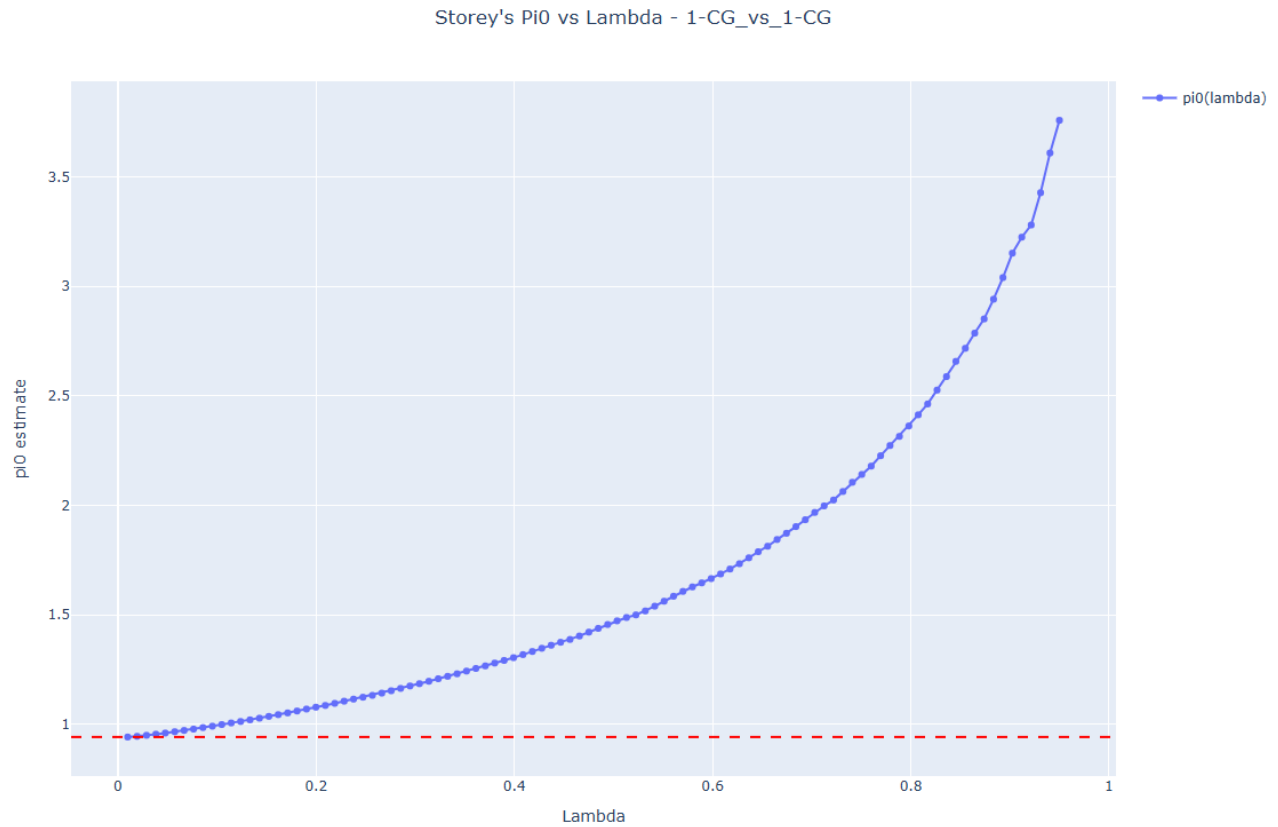
Potential future enhancements include adding adaptive significance thresholds, integrating covariates into the modeling framework, and extending support for additional statistical tests. For details on running analyses, see the `comparer.py` main function or the **MethylDetector** class implementation. Interpretation of results should consider potential dependencies between nearby genomic positions, since correlations can make global tests like Stouffer's more conservative and may reduce power.

References

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Appendix

Arabidopsis



Smart Centroid Comparison Summary

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Centroid 1: /home/ubuntu/Work/output_workflows/arabidopsis/centroids/WT/1-CG.h5

Centroid 2: /home/ubuntu/Work/output_workflows/arabidopsis/centroids/msh1/1-CG.h5

Total positions compared: 1331806

Significant positions: 80021

Significant fraction: 6.01%

Normal approximation used: 0

LRT used: 1331806

MoM fallbacks: 1798

FDR Analysis Results:

FDR method: storey

Pi0 estimate: 0.942

Global p-value (Stouffer): 0.000000
Global significant: True
Global threshold: 0.05
Mean q-value: 0.863772
Median q-value: 0.941934
Significant regions: 30601

Performance:

Processing time: 65.94 seconds
GPU acceleration: Yes
Significance level: 0.05

1-CG Statistics

- **total_positions** 1331806
- **significant_positions** 80021
- **significant_fraction** 0.06008457688281927
- **normal_approximation_count** 0
- **lrt_count** 1331806
- **fallback_to_mom_count** 1798
- **insufficient_sample_count** 12030
- **min_N** 10
- **mean_p_value** 0.663971315072652
- **median_p_value** 0.7808726454232446
- **mean_test_statistic** 3.214591695385574
- **median_test_statistic** 0.4946864169571086
- **min_p_value** 3.851863637256738e-39
- **max_p_value** 1
- **processing_time_seconds** 65.93977856636047
- **gpu_used** true
- **q_values_applied** true
- **fdr_method** "storey"
- **pi0_estimate** 0.9419342887580755
- **global_p_value** 0
- **global_z_score** -1313.774981642357
- **global_significant** true
- **global_significance_threshold** 0.05
- **significant_regions_count** 30601
- **mean_q_value** 0.8637717686078229
- **median_q_value** 0.9419342887580754
- **min_q_value** 3.419554715989113e-33

- **max_q_value** 0.9419342887580754